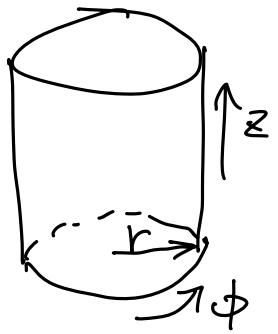
数理方程

第12周
20260515

柱函数

柱坐标下解PDE → 柱函数

$\nabla^2 u = 0$
- 球坐标：角向复杂，径向简单
- 柱坐标：角向简单，径向复杂 → Bessel



可理解为三角函数的"变形"（柱坐标径向不均匀带来的"变形"）

投石入水，水波以同心圆波形向外传播。

振幅随r减小，理想情况下振幅 $\sim \frac{1}{\sqrt{r}}$

Bessel函数特征

$x^2y''+xy'+(x^2-\nu^2)y=0$, $\nu\geq 0$ 为Bessel equation

求解过程见第8周ODE级数解

$$J_\nu(x)=\sum_{k=0}^{\infty}(-1)^k\frac{1}{k!\Gamma(\nu+k+1)}\left(\frac{x}{2}\right)^{2k+\nu}$$

$$J_{-\nu}(x)=\sum_{k=0}^{\infty}(-1)^k\frac{1}{k!\Gamma(-\nu+k+1)}\left(\frac{x}{2}\right)^{2k-\nu}$$

> 第一类柱函数 Bessel函数

$\nu\to n$ $(n=1,2,3\cdots)$

$\nu\to 0,-1,-2,\cdots$, $\Gamma(\nu)\to\infty$, 例: $\lim_{\nu\to 0}\Gamma(\nu)=\lim_{\nu\to 0}\frac{\Gamma(\nu+1)}{\nu}\sim\frac{1}{\nu}\to\infty$

$\Rightarrow J_{-\nu}(x)$前n项系数 $\frac{1}{\Gamma(-\nu+k+1)}\to 0$, $k=0,1,2\cdots,n-1$ ($\Gamma(-\nu+k+1)\to\infty$)

$$\Rightarrow J_{-n}(x)=\sum_{k=n}^{\infty}(-1)^k\frac{1}{k!\Gamma(-n+k+1)}\left(\frac{x}{2}\right)^{2k-n}$$

$$\overset{k'=k-n}{=}\sum_{k'=0}^{\infty}(-1)^{k'+n}\frac{1}{(k'+n)!\Gamma(k'+1)}\left(\frac{x}{2}\right)^{2k'+n}$$

$(k'+n)!=\Gamma(k'+n+1)$, $\Gamma(k'+1)=k'!$

$$=(-1)^n\sum_{k'=0}^{\infty}(-1)^{k'}\frac{1}{\Gamma(k'+n+1)\,k'!}\left(\frac{x}{2}\right)^{2k'+n}=(-1)^nJ_n(x)$$

$n=1,2,3\cdots$ J_{-n}, J_n不线性无关; $n=0$, $J_{-n}=J_n=J_0$

Neumann函数（第二类柱函数）

$$N_\nu(x)=\frac{\cos\nu\pi J_\nu(x)-J_{-\nu}(x)}{\sin\nu\pi},\qquad \{J_\nu(x),N_\nu(x)\}$$

二者独立（线性无关）

$\nu\to n\ (n=0,1,2\cdots)$ 分子、分母 $\to 0$

洛必塔法则

$$N_n(x)=\lim_{\nu\to n}\frac{\cos\nu\pi J_\nu(x)-J_{-\nu}(x)}{\sin\nu\pi}=\left.\frac{(-1)^n\frac{\partial J_\nu}{\partial\nu}-\frac{\partial J_{-\nu}}{\partial\nu}}{(-1)^n\pi}\right|_{\nu=n}$$

$$=\frac{1}{\pi}\left[\frac{\partial J_\nu}{\partial\nu}-(-1)^n\frac{\partial J_{-\nu}}{\partial\nu}\right]\Big|_{\nu=n}$$

$$=\frac{2}{\pi}J_n(x)\ln\frac{x}{2}+\cdots$$

↑Fuchs定理

$n=0\Rightarrow \frac{1}{\pi}\left[\frac{\partial J_\nu}{\partial\nu}+\frac{\partial J_\nu}{\partial(-\nu)}\right]_{\nu=0}=\frac{2}{\pi}\frac{\partial J_\nu}{\partial\nu}\Big|_{\nu=0}$

J_ν,N_ν 独立，$y(x)=C_1J_\nu(x)+C_2N_\nu(x)$

Hankel函数（第三类柱函数）$H_\nu^{(1)}=J_\nu+iN_\nu$，$H_\nu^{(2)}=J_\nu-iN_\nu$

例：$\int_0^\infty e^{-ax}J_0(bx)\,dx\quad(a>0)$

$$=\int_0^\infty e^{-ax}\sum_{k=0}^\infty\frac{(-1)^k}{k!\underbrace{\Gamma(k+1)}_{k!}}\left(\frac{bx}{2}\right)^{2k}dx\xrightarrow{t=ax}\sum_{k=0}^\infty\frac{(-1)^k}{(k!)^2}\left(\frac{b}{2}\right)^{2k}\frac{1}{a^{2k+1}}\int_0^\infty e^{-t}t^{2k}dt$$

$$=\sum_{k=0}^\infty\frac{(-1)^k}{(k!)^2}\left(\frac{b}{2}\right)^{2k}\frac{(2k)!}{a^{2k+1}}=\frac{1}{a}\sum_{k=0}^\infty\frac{1}{k!}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\cdots\left(-\frac{2k-1}{2}\right)\left(\frac{b}{a}\right)^{2k}$$

$$=\frac{1}{a}\left[1+\left(\frac{b}{a}\right)^2\right]^{-\frac{1}{2}}=\frac{1}{\sqrt{a^2+b^2}}$$

递推关系

$$g(x,t)=e^{\frac{x}{2}\left(t-\frac{1}{t}\right)}=\sum_{n=-\infty}^{\infty}J_n(x)t^n$$

① $\frac{d}{dx}[x^\nu J_\nu(x)]=x^\nu J_{\nu-1}(x)$

② $\frac{d}{dx}[x^{-\nu}J_\nu(x)]=-x^{-\nu}J_{\nu+1}(x)$

> 基本递推关系

① 证明: $\frac{d}{dx}[x^\nu J_\nu] = \frac{d}{dx}\left[\sum_{k=0}^{\infty}(-1)^k \frac{1}{k!\Gamma(k+\nu+1)}\frac{1}{2^{2k+\nu}}x^{2k+2\nu}\right]$

$= \sum_{k=0}^{\infty}\frac{(-1)^k}{k!\Gamma(\nu+k+1)}\frac{2(k+\nu)}{2^{2k+\nu}}x^{2k+2\nu-1} = \sum_{k=0}^{\infty}(-1)^k\frac{1}{k!\Gamma(\nu+k)}\left(\frac{x}{2}\right)^{2k+\nu-1}x^\nu$

$\frac{k+\nu}{\Gamma(\nu+k+1)} = \frac{1}{\Gamma(\nu+k)}$

$= x^\nu J_{\nu-1}(x)$

② 类似

$N_\nu(x)$ 满足类似关系 $\quad \frac{d}{dx}[x^\nu N_\nu(x)] = x^\nu N_{\nu-1}(x)$,

$\frac{d}{dx}[x^{-\nu}N_\nu(x)] = -x^{-\nu}N_{\nu+1}(x)$

①② $\Rightarrow$ ③ $J_{\nu-1}(x) - J_{\nu+1}(x) = 2J_\nu'(x)$

④ $J_{\nu-1}(x) + J_{\nu+1}(x) = \frac{2\nu}{x}J_\nu(x)$

证明:

① $\Rightarrow x^{-\nu}\frac{d}{dx}[x^\nu J_\nu] = J_{\nu-1} \Rightarrow \frac{\nu}{x}J_\nu + J_\nu' = J_{\nu-1}$ (A)

② $\Rightarrow x^\nu\frac{d}{dx}[x^{-\nu}J_\nu] = -J_{\nu+1} \Rightarrow -\frac{\nu}{x}J_\nu + J_\nu' = -J_{\nu+1}$ (B)

(A)+(B) $\Rightarrow$ ③, (A)−(B) $\Rightarrow$ ④

应用: 幂函数乘 Bessel 积分.

$\int_0^1 (1-x^2)J_0(\mu x)\,x\,dx$ μ 是 $J_0(x)$ 的零点. $J_0(\mu)=0$

① $\Rightarrow \frac{d}{d(\mu x)}[\mu x J_1(\mu x)] = \mu x J_0(\mu x) \Rightarrow \frac{1}{\mu}\frac{d}{dx}[xJ_1(\mu x)] = xJ_0(\mu x)$

原积分 $= \int_0^1 (1-x^2)\frac{1}{\mu}\frac{d}{dx}[xJ_1(\mu x)]\,dx$

$= (1-x^2)\frac{1}{\mu}xJ_1(\mu x)\Big|_0^1 + \frac{2}{\mu}\int_0^1 x^2 J_1(\mu x)\,dx$

$(1-x^2)\frac{1}{\mu}xJ_1(\mu x)\Big|_0^1 = 0$

$\frac{d}{d(\mu x)}[(\mu x)^2 J_2(\mu x)] = (\mu x)^2 J_1(\mu x)$

$= \frac{2}{\mu^2}\int_0^1 \frac{d}{dx}[x^2 J_2(\mu x)]\,dx = \frac{2}{\mu^2}x^2 J_2(\mu x)\Big|_0^1 = \frac{2}{\mu^2}J_2(\mu)$

$$\text{①} \Rightarrow \underbrace{J_0(\mu)}_{=0} + J_2(\mu) = \frac{2}{\mu} J_1(\mu) \Rightarrow J_2(\mu) = \frac{2}{\mu} J_1(\mu)$$

$$\Rightarrow \text{同理} = \frac{2}{\mu^2} J_2(\mu) = \frac{4}{\mu^3} J_1(\mu)$$

$$\int_a^b x^\mu J_\nu(x)\,dx = \int_a^b x^{\mu-\nu-1} x^{\nu+1} J_\nu\,dx = x^{\mu-\nu-1} x^{\nu+1} J_{\nu+1}\Big|_a^b - (\mu-\nu-1)\int_a^b x^{\mu-\nu-2} x^{\nu+1} J_{\nu+1}\,dx$$

$$= \frac{d}{dx}\left[x^{\nu+1} J_{\nu+1}\right]$$

$$= x^\mu J_{\nu+1}\Big|_a^b - (\mu-\nu-1)\int_a^b x^{\mu-\nu-3} x^{\nu+2} J_{\nu+1}\,dx$$

$$x^{\mu-1} \qquad \frac{d}{dx}\left[x^{\nu+2} J_{\nu+2}\right]$$

$$\Downarrow$$

$$\int_a^b x^{\mu-n} J_{\nu+n}\,dx$$

两类分析情况 $\begin{cases} (\mu-n)-(\nu+n)=1, & \mu-\nu=2n+1 \\ (\mu-n)+(\nu+n)=1 & \mu+\nu=1 \end{cases}$

$$\int_a^b x^{\nu+n+1} J_{\nu+n}\,dx \overset{①}{=} \int_a^b \frac{d}{dx}\left[x^{\nu+n+1} J_{\nu+n+1}\right] dx$$

$$\int_a^b x^{-(\nu+n-1)} J_{\nu+n}\,dx \overset{②}{=} -\int_a^b \frac{d}{dx}\left[x^{-(\nu+n-1)} J_{\nu+n-1}\right] dx$$

渐近行为：$x\to 0$，$x\to\infty$ 是否发散？ $\nu\geq 0$

A) $x\to 0$ $\quad J_\nu(x)=\sum_{k=0}^{\infty}(-1)^k\frac{1}{k!\Gamma(\nu+k+1)}\left(\frac{x}{2}\right)^{2k+\nu}\sim\frac{1}{\Gamma(\nu+1)}\left(\frac{x}{2}\right)^{\nu}$ （$k=0$ 主导）

$$\begin{cases}\nu=0 & J_0(x)\to 1\\ \nu>0 & J_\nu(x)\sim\frac{1}{\Gamma(\nu+1)}\left(\frac{x}{2}\right)^{\nu}\to 0\end{cases}$$

$\nu>0$：$N_\nu(x)=\dfrac{\cos\nu\pi\,J_\nu(x)-J_{-\nu}(x)}{\sin\nu\pi}\sim-\dfrac{J_{-\nu}(x)}{\sin\nu\pi}\overset{k=0\text{主导}}{\sim}-\dfrac{1}{\sin\nu\pi}\dfrac{1}{\Gamma(1-\nu)}\left(\dfrac{x}{2}\right)^{-\nu}$

$\Gamma(x)\Gamma(1-x)=\dfrac{\pi}{\sin\pi x}$

$$\sim-\frac{\Gamma(\nu)\Gamma(1-\nu)}{\pi}\frac{1}{\Gamma(1-\nu)}\left(\frac{x}{2}\right)^{-\nu}$$

$$\sim-\frac{\Gamma(\nu)}{\pi}\left(\frac{x}{2}\right)^{-\nu}$$

$\nu=0$，$N_0(x)\sim\dfrac{2}{\pi}J_0(x)\ln\dfrac{x}{2}+$ 幂级数

$\sim\dfrac{2}{\pi}\ln\dfrac{x}{2}$

即 $\begin{cases}\nu=0 & N_0(x)\sim\frac{2}{\pi}\ln\frac{x}{2}\to-\infty\\ \nu>0 & N_\nu(x)\sim-\frac{\Gamma(\nu)}{\pi}\left(\frac{x}{2}\right)^{-\nu}\to-\infty\end{cases}$ > 发散！

B) $x\to\infty$ $\quad J_\nu(x)\sim\sqrt{\frac{2}{\pi x}}\cos(x-\frac{\nu\pi}{2}-\frac{\pi}{4})$ $\quad$ 振荡衰减

$N_\nu(x)\sim\sqrt{\frac{2}{\pi x}}\sin(x-\frac{\nu\pi}{2}-\frac{\pi}{4})$

0阶近似 $\quad y''+\frac{1}{x}y'+(1-\frac{\nu^2}{x^2})y=0 \Rightarrow y''+y\approx 0$

$\Rightarrow y(x)\sim\cos(x-x_0)$

↑使得 $y''+\frac{1}{x}y'+(1-\frac{\nu^2}{x^2})y\sim O(x^{-1})$，↑误差量级

一阶近似 $\quad y(x)=f(x)\cos(x-x_0)$ 求$f(x)$使得方程以更高精度满足

$$\underbrace{(f''\cos-2f'\sin-f\cos)}_{y''}+\frac{1}{x}\underbrace{(f'\cos-f\sin)}_{y'}+(1-\frac{\nu^2}{x^2})\underbrace{f\cos}_{y}=0$$

$$\Rightarrow (f''\cos-2f'\sin)+\frac{1}{x}(f'\cos-f\sin)-\frac{\nu^2}{x^2}f\cos=0$$

$$-(2f'+\frac{f}{x})\sin x+(f''+\frac{1}{x}f'-\frac{\nu^2}{x^2}f)\cos x=0$$

↑令sin系数$=0$ 扔掉

$\Rightarrow 2f'+\frac{f}{x}=0 \Rightarrow f(x)\sim x^{-\frac{1}{2}}$

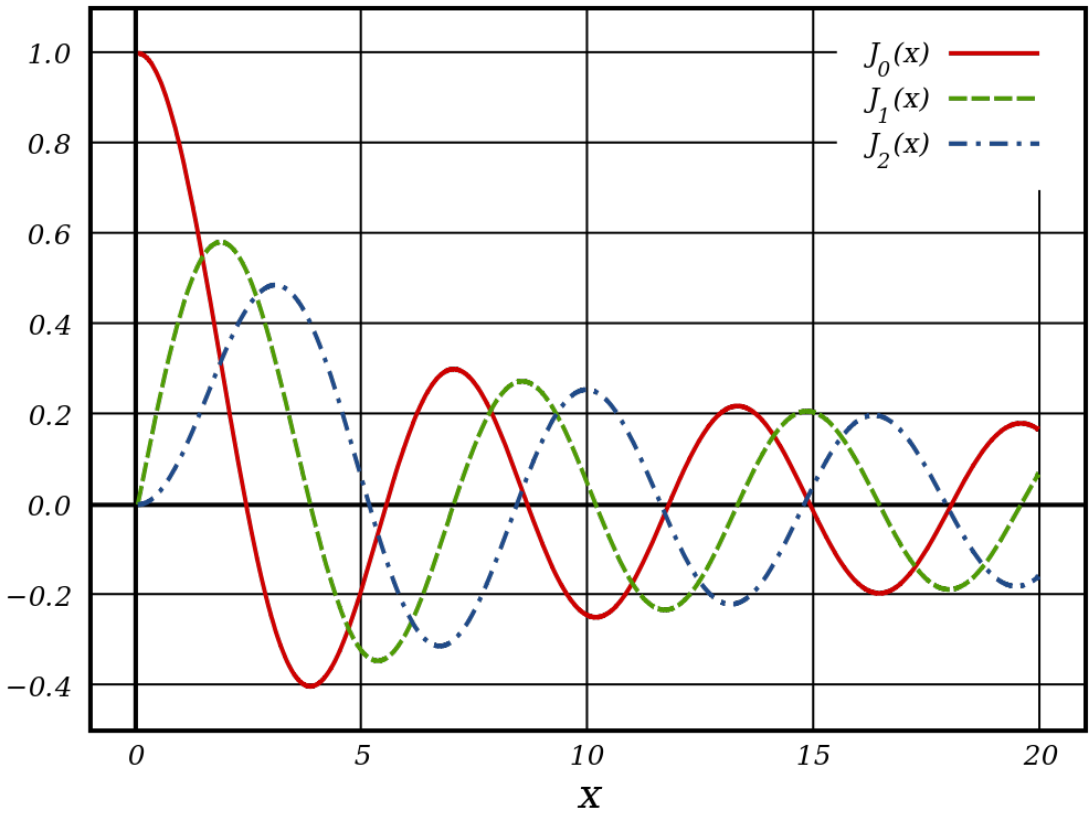
cos系数 $f''+\frac{1}{x}f'-\frac{\nu^2}{x^2}f\sim x^{-\frac{5}{2}}$

$\Rightarrow y(x)=f(x)\cos(x-x_0)=x^{-\frac{1}{2}}\cos(x-x_0)$使得 $y''+\frac{1}{x}y'+(1-\frac{\nu^2}{x^2})y\sim O(x^{-\frac{5}{2}})$

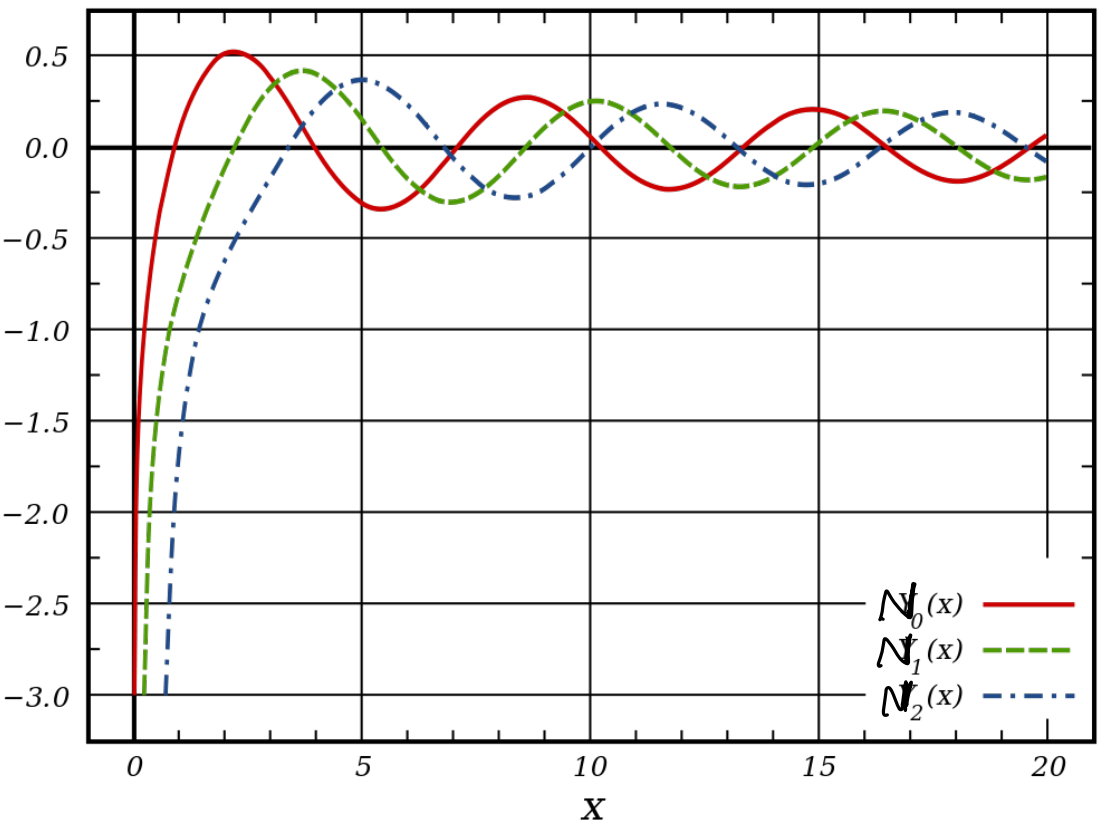
考虑到 $y(x)=O(x^{-\frac{1}{2}})$，$x^{-\frac{5}{2}}/x^{-\frac{1}{2}}\sim x^{-2}$，误差量级由之前的$O(x^{-1})$减小至$O(x^{-2})$.

$$y(x)=f(x)\cos(x-x_0)=\sqrt{\frac{1}{x}}\cos(x-x_0)$$，振荡衰减

Bessel函数

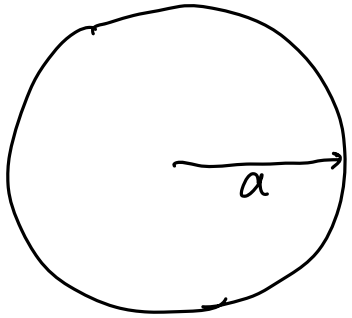


Neumann函数



应用：$\frac{\partial u}{\partial t}=D\nabla^2 u,\quad r<a$

$u|_{r=a}=0,\quad u|_{t=0}=f(r,\phi)$



$u|_{\phi=0}=u|_{\phi=2\pi},\quad \frac{\partial u}{\partial \phi}\Big|_{\phi=0}=\frac{\partial u}{\partial \phi}\Big|_{\phi=2\pi}$

与矩形做法类似：

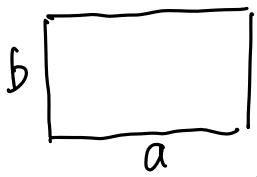
$\frac{\partial u}{\partial t}=D\nabla^2 u\quad 0<x<a,\ 0<y<b$

$u|_{x=0}=u|_{x=a}=u|_{y=0}=u|_{y=b}=0$

$u|_{t=0}=f(x,y)$

$\Rightarrow u(x,y,t)=\sum_{m=1}^{\infty}\sum_{n=1}^{\infty} A_{mn}\sin\frac{m\pi}{a}x\sin\frac{n\pi}{b}y\, e^{-D[(\frac{m\pi}{a})^2+(\frac{n\pi}{b})^2]t}$

$V(x,y)$　　$T(t)$



对圆盘：

$u=V(r,\phi)T(t)\Rightarrow \frac{\nabla^2 V}{V}=\frac{T'}{DT}=-E$ ← 本征值

$\nabla^2 V+EV=0,\qquad T'=-DET$

$V|_{r=a}=0,\quad V|_{r=0}<\infty$

$V|_{\phi=0}=V|_{\phi=2\pi},\quad \frac{\partial V}{\partial\phi}\Big|_{\phi=0}=\frac{\partial V}{\partial\phi}\Big|_{\phi=2\pi}$

$\frac{1}{r}\frac{\partial}{\partial r}(r\frac{\partial V}{\partial r})+\frac{1}{r^2}\frac{\partial^2 V}{\partial\phi^2}+EV=0$

$\frac{1}{r}\frac{\partial}{\partial r}(r\frac{\partial R}{\partial r})\Phi$　　$-\frac{1}{r^2}R\lambda\Phi$　　$ER\Phi$

除以 Φ

$v(r,\phi)=R(r)\Phi(\phi)\Rightarrow$

$\Phi''+\lambda\Phi=0$

$\Phi(0)=\Phi(2\pi),\quad \Phi'(0)=\Phi'(2\pi)$

$\frac{1}{r}\frac{d}{dr}(r\frac{dR}{dr})+(E-\frac{\lambda}{r^2})R=0$

$R(0)<\infty,\quad R(a)=0$

角向：$\lambda=m^2$，$\Phi_m=\begin{cases}\cos m\phi & m=0,1,2\cdots\\ \sin m\phi & m=1,2\cdots\end{cases}$

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{dR}{dr}\right)+\left(k^2-\frac{\lambda}{r^2}\right)R=0$$

↑记 $E=k^2$

从S-L理论，易得 $E>0$.

根据后续计算也可知 $E>0$. 若 $E<0$，将出现虚宗量 Bessel函数 $J_m(i\sqrt{|E|}\,r)$，行为类似指数函数，无零点，无法满足 $R(a)=0$.

$$r\frac{d}{dr}\left(r\frac{dR}{dr}\right)+(k^2r^2-m^2)R=0$$

$x=kr$ ⟹ $$x\frac{d}{dx}\left(x\frac{dR}{dx}\right)+(x^2-m^2)R=0$$

$$x^2R''+xR'+(x^2-m^2)R=0$$

$$\Rightarrow R=CJ_m(x)+\cancel{DN_m(x)}=CJ_m(x)=CJ_m(kr)$$

$R(0)<\infty$

$$R(a)=0\Rightarrow J_m(ka)=0\Rightarrow ka=\mu_i^{(m)}$$

↑ $J_m(x)$ 第 i 个正零点.

$$\Rightarrow k_{mi}=\frac{\mu_i^{(m)}}{a}$$

本征值 $E_{mi}=\left(\frac{\mu_i^{(m)}}{a}\right)^2$，$m=0,1,2\cdots$，$i=1,2,3\cdots$

径向本征函数 $R_{mi}(r)=J_m(k_{mi}r)=J_m\left(\frac{\mu_i^{(m)}}{a}r\right)$

v 的本征函数 $\begin{cases}v_{mi1}=R_{mi}(r)\cos m\phi=J_m\left(\frac{\mu_i^{(m)}}{a}r\right)\cos m\phi & m=0,1,2,\ i=1,2,\cdots\\ v_{mi2}=R_{mi}(r)\sin m\phi=J_m\left(\frac{\mu_i^{(m)}}{a}r\right)\sin m\phi & m=1,2,3,\ i=1,2\cdots\end{cases}$

$$u(r,\phi,t)=\left(\sum_m\sum_i A_{mi}v_{mi1}+\sum_m\sum_i B_{mi}v_{mi2}\right)e^{-DE_{mi}t}$$

$$=\left[\sum_{m=0}^{\infty}\sum_{i=1}^{\infty}A_{mi}R_{mi}(r)\cos m\phi+\sum_{m=1}^{\infty}\sum_{i=1}^{\infty}B_{mi}R_{mi}(r)\sin m\phi\right]e^{-DE_{mi}t}$$

r,ϕ 二重级数

$$u|_{t=0} = f(r,\phi) = \sum_{m=0}^{\infty}\sum_{i=1}^{\infty} A_{mi} R_{mi}(r)\cos m\phi + \sum_{m=1}^{\infty}\sum_{i=1}^{\infty} B_{mi} R_{mi}(r)\sin m\phi$$

$$f(r,\phi) = \sum_{m=0}^{\infty} f_{m1}(r)\cos m\phi + \sum_{m=1}^{\infty} f_{m2}(r)\sin m\phi \quad \leftarrow f(r,\phi)\text{傅里叶展开}$$

$$f_{m1}(r) = \frac{1}{\pi(1+\delta_{m0})}\int_0^{2\pi} f(r,\phi)\cos m\phi\, d\phi \qquad f_{m2}(r) = \frac{1}{\pi}\int_0^{2\pi} f(r,\phi)\sin m\phi\, d\phi$$

$$\Rightarrow f_{m1}(r) = \sum_{i=1}^{\infty} A_{mi} R_{mi}(r), \quad = \sum_{i=1}^{\infty} A_{mi} J_m\left(\frac{\mu_i^{(m)}}{a} r\right)$$

$$f_{m2}(r) = \sum_{i=1}^{\infty} B_{mi} R_{mi}(r) = \sum_{i=1}^{\infty} B_{mi} J_m\left(\frac{\mu_i^{(m)}}{a} r\right)$$

即 $f_{m1}(r)$, $f_{m2}(r)$ 按 $J_m\left(\frac{\mu_i^{(m)}}{a} r\right)$ 展开.

由正交性定系数 A_{mi}.

方程改写为S-L标准形式：（本征值 λ，权函数 r）

$$\frac{d}{dr}\left(r\frac{dR}{dr}\right) + \left(\lambda r - \frac{m^2}{r}\right)R = 0$$

$$R(0)<\infty, \quad R(a)=0$$

$$\Rightarrow \int_0^a J_m\left(\frac{\mu_i^{(m)}}{a} r\right) J_m\left(\frac{\mu_j^{(m)}}{a} r\right) r\, dr = 0 \quad (i \neq j)$$

广义傅里叶级数：

$$[0,a]: \quad f(r) = \sum_{i=1}^{\infty} b_i J_m\left(\frac{\mu_i^{(m)}}{a} r\right)$$

$$b_i = \frac{1}{\left\| J_m\left(\frac{\mu_i^{(m)}}{a} r\right)\right\|^2} \int_0^a f(r) J_m\left(\frac{\mu_i^{(m)}}{a} r\right) r\, dr$$

↑ 权函数

$$\|J_m(\frac{\mu_i^{(m)}}{a}r)\|^2 = \int_0^a [J_m(\frac{\mu_i^{(m)}}{a}r)]^2 r\,dr \xrightarrow{x=\frac{\mu_i^{(m)}}{a}r} (\frac{a}{\mu_i^{(m)}})^2 \int_0^{\mu_i^{(m)}} [J_m(x)]^2 x\,dx$$

$$= \frac{a^2}{2}[J_{m+1}(\mu_i^{(m)})]^2$$

$$\int_0^a [J_m(x)]^2 x\,dx = \frac{a^2}{2}[J_m'(a)]^2 + \frac{1}{2}(a^2-m^2)[J_m(a)]^2, \quad m \geq 0$$

若 $J_m(a)=0$ $\Rightarrow \frac{a^2}{2}[J_{m+1}(a)]^2$

推导: $x(xJ_m')' + (x^2-m^2)J_m = 0$

乘 $2J_m'$ $\Rightarrow 2xJ_m'(xJ_m')' + 2(x^2-m^2)J_mJ_m' = 0$

$\Rightarrow [(xJ_m')^2]' + (x^2-m^2)(J_m^2)' = 0$

$\Rightarrow [(xJ_m')^2 + (x^2-m^2)J_m^2]' = 2xJ_m^2$

$\int_0^a$ $\Rightarrow \int_0^a x[J_m(x)]^2 dx = \frac{1}{2}[(xJ_m')^2 + (x^2-m^2)J_m^2]\Big|_0^a$

$= \frac{a^2}{2}[J_m'(a)]^2 + \frac{1}{2}(a^2-m^2)[J_m(a)]^2$

$(x^2-m^2)J_m^2|_{x=0} = 0$

$m>0$: $J_m|_{x=0} = 0$

$m=0$: $(x^2-m^2)_{x=0} = 0$

进一步, 如果 $J_m(a)=0$,

$\Rightarrow \int_0^a x[J_m(x)]^2 dx = \frac{a^2}{2}[J_m'(a)]^2$

递推关系② $\Rightarrow \frac{d}{dx}[x^{-m}J_m(x)] = -x^{-m}J_{m+1}(x) \Rightarrow -m\frac{J_m}{x} + J_m' = -J_{m+1}(x)$

$\Downarrow x=a,\ J_m(a)=0$

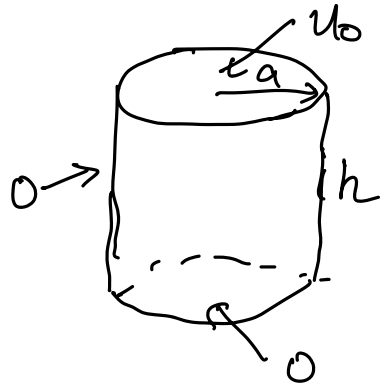
$J_m'(a) = -J_{m+1}(a)$

$\Downarrow$

$\int_0^a x[J_m(x)]^2 dx = \frac{a^2}{2}[J_{m+1}(a)]^2$

可进一步化简

例：圆柱体内稳定温度分布



$\nabla^2 u = 0 \quad r<a,\ 0<z<h$

$u|_{r=a}=0,\ u|_{z=0}=0,\ u|_{z=h}=u_0$

$\Rightarrow u(r,z)$

$\nabla^2 u = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right)+\frac{\partial^2 u}{\partial z^2}+\cancel{\frac{1}{r^2}\frac{\partial^2 u}{\partial \varphi^2}}=0$

$u(r,z)=R(r)Z(z)$

$\Downarrow$

$(rR')'+\lambda rR=0 \qquad Z''-\lambda Z=0$

$\langle R(0)<\infty,\ R(a)=0$

$r^2R''+rR'+\lambda r^2R=0 \xrightarrow{x=\sqrt{\lambda}r} x^2\frac{d^2R}{dx^2}+x\frac{dR}{dx}+(x^2-0^2)R=0$

$\uparrow \ \nu=0$

$\Rightarrow R=J_0(x)=J_0(\sqrt{\lambda}r)$

N₀舍去 ← $\mu_i=\mu_i^{(0)}$，J_0零点

$R(a)=0 \Rightarrow J_0(\sqrt{\lambda}a)=0 \Rightarrow \lambda_i=\left(\frac{\mu_i}{a}\right)^2$

$R_i(r)=J_0\left(\frac{\mu_i}{a}r\right) \quad i=1,2,3\cdots$

$\Rightarrow Z_i(z)=A_ie^{\frac{\mu_i}{a}z}+B_ie^{-\frac{\mu_i}{a}z}$

$u|_{z=0}=0 \Rightarrow A_i+B_i=0$

$\Rightarrow u=\sum_{i=1}^{\infty}R_i(r)Z_i(z)=\sum_{i=1}^{\infty}C_i\sinh\left(\frac{\mu_i}{a}z\right)J_0\left(\frac{\mu_i}{a}r\right)$

$u|_{z=h}=u_0=\sum_{i=1}^{\infty}C_i\sinh\frac{\mu_i h}{a}J_0\left(\frac{\mu_i}{a}r\right)$

$$\Rightarrow C_i \sinh\frac{\mu_i h}{a} = \frac{1}{\|J_0(\frac{\mu_i}{a}r)\|^2}\int_0^a u_0 J_0(\frac{\mu_i}{a}r)\, r\,dr.$$

$$\|J_0\|^2 = \int_0^a [J_0(\frac{\mu_i}{a}r)]^2 r\,dr = \frac{a^2}{2}[J_1(\mu_i)]^2 \quad (\text{调用前面结果})$$

$$\int_0^a u_0 J_0(\frac{\mu_i}{a}r)\, r\,dr \overset{x=\frac{\mu_i}{a}r}{=\!=} u_0(\frac{a}{\mu_i})^2\int_0^{\mu_i} J_0(x)\,x\,dx$$

$$= u_0(\frac{a}{\mu_i})^2\int_0^{\mu_i}\frac{d}{dx}(xJ_1)\,dx = u_0(\frac{a}{\mu_i})^2\mu_i J_1(\mu_i) = u_0\frac{a^2}{\mu_i}J_1(\mu_i)$$

$$\frac{d}{dx}(x^\nu J_\nu) = x^\nu J_{\nu-1}$$

$$= \frac{u_0\frac{a^2}{\mu_i}J_1(\mu_i)}{\frac{a^2}{2}[J_1(\mu_i)]^2} = \frac{2u_0}{\mu_i J_1(\mu_i)}$$

$$\Rightarrow C_i = \frac{2u_0}{\mu_i J_1(\mu_i)}\frac{1}{\sinh(\frac{\mu_i h}{a})}$$

$$\Rightarrow u = \sum_{i=1}^{\infty}\frac{2u_0}{\mu_i}\frac{\sinh(\frac{\mu_i}{a}z)}{\sinh(\frac{\mu_i}{a}h)}\frac{J_0(\frac{\mu_i}{a}r)}{J_1(\mu_i)}$$

虚宗量Bessel函数

J_ν, N_ν 类似 sin, cos

$\sin x, \cos x$

$\downarrow$

$\sin(ix) = \frac{e^{i(ix)} - e^{-i(ix)}}{2i} = \frac{e^{-x} - e^{x}}{2i}$

$\cos(ix) = \cdots$

$J_\nu(x), N_\nu(x)$

$\downarrow$

$J_\nu(ix) = ?$ ← 性质类似于 $e^{\pm x}$

$N_\nu(ix) = ?$

$$J_\nu(ix) = J_\nu\left(e^{\frac{\pi}{2}i}x\right) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\Gamma(k+\nu+1)} \left(\frac{x}{2}e^{\frac{\pi}{2}i}\right)^{2k+\nu}$$

$$= e^{\frac{\pi\nu}{2}i} \sum_{k=0}^{\infty} \frac{(-1)^k e^{k\pi i}}{k!\Gamma(k+\nu+1)} \left(\frac{x}{2}\right)^{2k+\nu} \qquad (-1)^k$$

$$= e^{\frac{\pi\nu}{2}i} \sum_{k=0}^{\infty} \frac{1}{k!\Gamma(k+\nu+1)} \left(\frac{x}{2}\right)^{2k+\nu}$$

$$I_\nu(x) = e^{-\frac{\pi\nu}{2}i} J_\nu(ix) = \sum_{k=0}^{\infty} \frac{1}{k!\Gamma(k+\nu+1)} \left(\frac{x}{2}\right)^{2k+\nu}$$

↑

第一类虚宗量Bessel函数 (Modified Bessel function)

$J_\nu(ix)$ 满足 $(ix)^2\frac{d^2J_\nu(ix)}{d(ix)^2}+ix\frac{dJ_\nu(ix)}{d(ix)}+[(ix)^2-\nu^2]J_\nu(ix)=0$

$\Rightarrow x^2\frac{d^2J_\nu(ix)}{dx^2}+x\frac{dJ(ix)}{dx}+(-x^2-\nu^2)J_\nu(ix)=0$

$I_\nu(x)\propto J_\nu(ix)\Rightarrow I_\nu(x)$ 满足同样的ODE.

直接求级数解也可得到 $I_\nu=\sum_{k=0}^{\infty}(\cdots)$

$$x^2I_\nu''(x)+xI_\nu'(x)+(-x^2-\nu^2)I_\nu(x)=0$$

与Bessel equation差负号

Modified Bessel equation

$x\to\infty$，作最粗糙近似，

$\Rightarrow x^2I_\nu''-x^2I_\nu\approx0\Rightarrow I_\nu''-I_\nu\approx0\Rightarrow I_\nu(x)\sim e^{\pm x}$

$I_\nu(x)$行为类似于指数函数

$I_{-n}(x)=I_n(x)\quad n=0,1,2\cdots$

第二类虚宗量Bessel函数：与$N_\nu(x)$定义方式类似

$K_\nu(x)=\frac{\pi}{2\sin\nu\pi}[I_{-\nu}(x)-I_\nu(x)]$，与$I_\nu(x)$线性无关

$K_n(x)=\lim_{\nu\to n}K_\nu(x)\quad(n=0,1,2\cdots)$

↑这种方法来计算

I_ν, K_ν, $x\to\infty$ 渐近行为可由 $J_{\pm\nu}$ 得出：

$$I_\nu(x) = e^{-\frac{\pi\nu i}{2}} J_\nu(ix) \sim e^{-\frac{\pi\nu}{2}i}\sqrt{\frac{2}{\pi i x}}\cos\left(ix-\frac{\nu\pi}{2}-\frac{\pi}{4}\right)$$

$$\sim e^{-\frac{\pi\nu i}{2}}\sqrt{\frac{2}{\pi x}}\, e^{-\frac{\pi i}{4}}\,\frac{1}{2}\left[e^{-i\left(ix-\frac{\nu\pi}{2}-\frac{\pi}{4}\right)} + e^{i\left(ix-\frac{\nu\pi}{2}-\frac{\pi}{4}\right)}\right]$$

$$\frac{1}{\sqrt{i}} = e^{-\frac{\pi i}{4}}$$

$$\sim \frac{1}{\sqrt{2\pi x}} e^{x}$$

主要特征：

$x\to 0$：$I_0(x)\to 1$，$I_1(x), I_2(x), I_3(x)\ldots \to 0$

$K_0(x), K_1(x), K_2(x), \ldots \to \infty$

$x\to\infty$：$I_\nu(x) \sim \sqrt{\frac{1}{2\pi x}}\, e^{x}$

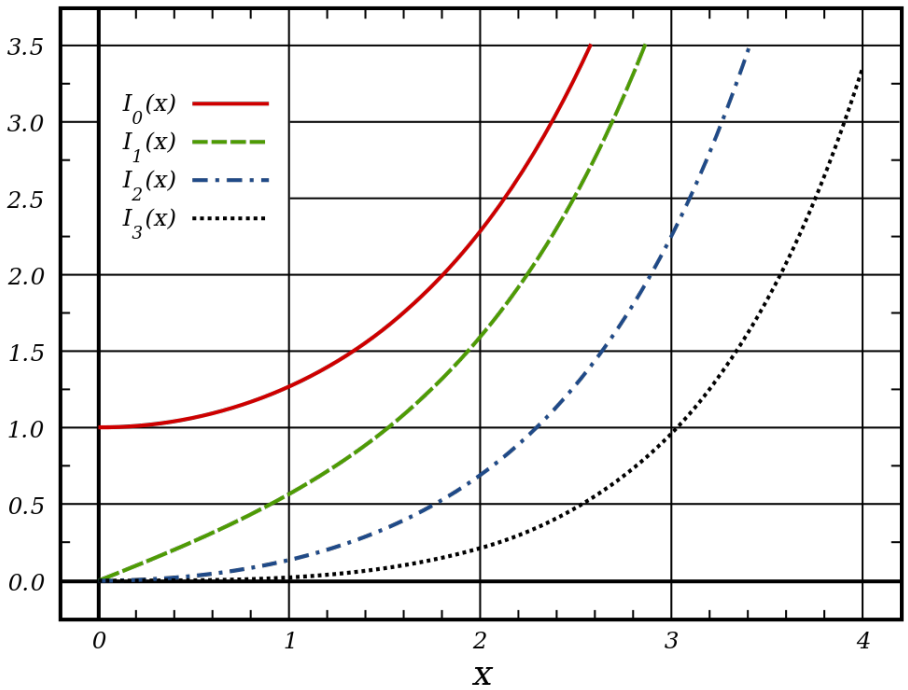
$K_\nu(x) \sim \sqrt{\frac{1}{2x}}\, e^{-x}$

$\sin x, \cos x$ $\xrightarrow{x\to ix}$ $e^{\pm x}$

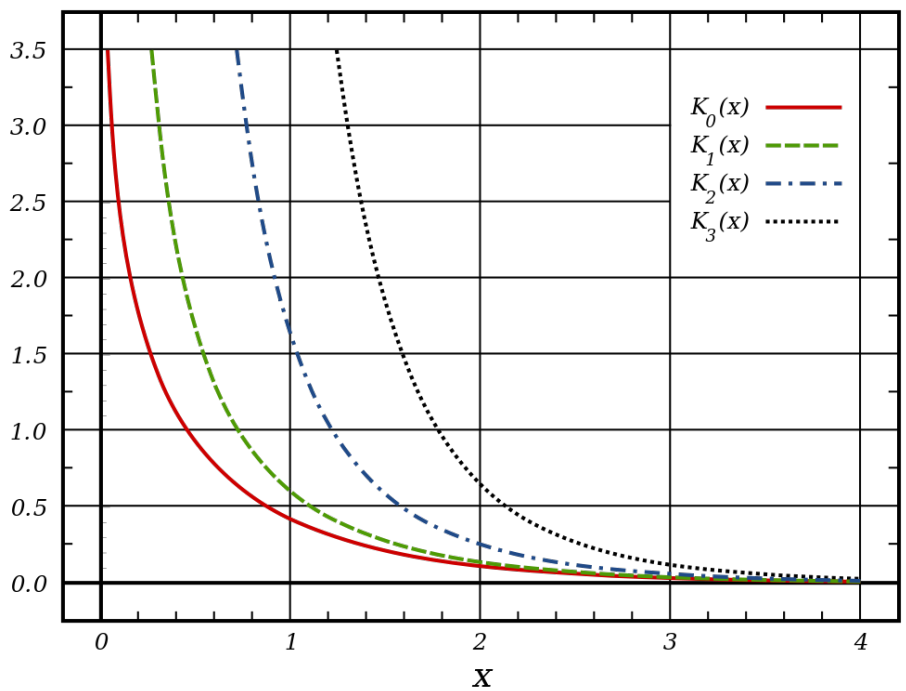
$J_{\pm\nu}(x)$ $\xrightarrow{x\to ix}$ $I_\nu(x), K_\nu(x)$

$\frac{1}{\sqrt{x}} e^{\pm x}$

第一类

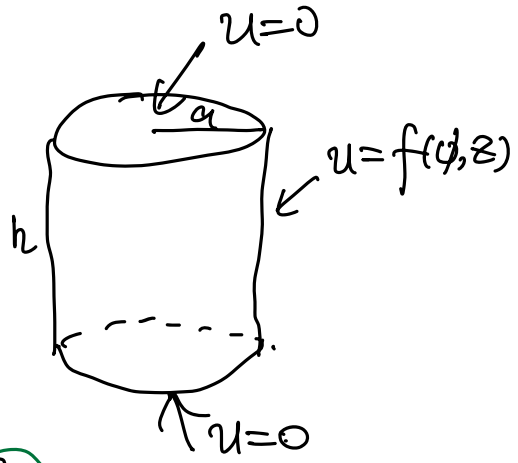

3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
1
2
3
4
X
$I_0(x)$
$I_1(x)$
$I_2(x)$
$I_3(x)$

第二类


3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
1
2
3
4
X
$K_0(x)$
$K_1(x)$
$K_2(x)$
$K_3(x)$

应用

$$\nabla^2 u = 0 \quad r<a,\ 0<z<h$$
$$u|_{r=0} < \infty,\quad u|_{r=a} = f(\phi, z)$$
$$u|_{z=0} = u|_{z=h} = 0$$



$$u|_{\phi=0} = u|_{\phi=2\pi},\quad \left.\frac{\partial u}{\partial \phi}\right|_{\phi=0} = \left.\frac{\partial u}{\partial \phi}\right|_{\phi=2\pi}$$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \phi^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

$\frac{\partial^2 u}{\partial \phi^2} \to -\mu R\Phi Z$, $\frac{\partial^2 u}{\partial z^2} \to -\lambda R\Phi Z$

$$u = R(r)\Phi(\phi)Z(r)$$

$$\Phi'' + \mu\Phi = 0,\quad \Phi(0) = \Phi(2\pi),\quad \Phi'(0) = \Phi'(2\pi)$$ ← 本征值问题

$$Z'' + \lambda Z = 0,\quad Z(0) = Z(h) = 0$$ ←

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{dR}{dr}\right) + \left(-\lambda - \frac{\mu}{r^2}\right)R = 0$$

Φ: $\Phi_{m1} = \cos m\phi,\quad \Phi_{m2} = \sin m\phi$

Z: $\lambda_n = \left(\frac{n\pi}{h}\right)^2 \quad n = 1, 2\cdots \quad Z_n = \sin\frac{n\pi}{h}z$

$$\Rightarrow r\frac{d}{dr}\left(r\frac{dR}{dr}\right) + \left[-\left(\frac{n\pi}{h}r\right)^2 - m^2\right]R = 0$$

$x = \frac{n\pi}{h}r$ ⟹

$$x\frac{d}{dx}\left(x\frac{dR}{dx}\right) + \left[-x^2 - m^2\right]R = 0$$

$$R = C I_m(x) + D K_m(x)$$ ~~$D K_m(x)$~~ $R(0) \to \infty$

$$= I_m\left(\frac{n\pi}{h}r\right)$$

$$\Rightarrow u = \sum_{m=0}^{\infty}\sum_{n=1}^{\infty}\left(A_{mn}\cos m\phi + B_{mn}\sin m\phi\right) I_m\left(\frac{n\pi}{h}r\right)\sin\frac{n\pi}{h}z$$

$$u|_{r=a} = f(\phi, z)$$

$$= \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} (A_{mn}\cos m\phi + B_{mn}\sin m\phi)\, I_m\left(\frac{n\pi}{h}a\right) \sin\frac{n\pi}{h}z$$

正交性:

$$A_{mn} I_m\left(\frac{n\pi}{h}a\right) = \frac{1}{\pi}\frac{1}{1+\delta_{m0}}\frac{2}{h}\int_0^{2\pi} d\phi \int_0^h dz\, f(\phi, z)\cos m\phi \sin\frac{n\pi}{h}z$$

$$B_{mn} I_m\left(\frac{n\pi}{h}a\right) = \frac{1}{\pi}\frac{2}{h}\int_0^{2\pi} d\phi \int_0^h dz\, f(\phi, z)\sin m\phi \sin\frac{n\pi}{h}z$$

本题中 ϕ, z 方向本征函数为三角函数, 自然期待 r 方向函数行为类似指数函数, 结果确实如此. $I_m\left(\frac{n\pi}{h}r\right)$ 在 r 较大时形似指数函数.

回忆长方体中的 Laplace 方程, 若两个方向为三角函数, 第三个方向函数行为是指数函数.

